

Neural Networks are taking over!

- Neural networks have become one of *the* main approaches to AI
- They have been successfully applied to various pattern recognition, prediction, and analysis problems
- In many problems they have established the state of the art
 - Often exceeding previous benchmarks by large margins
 - Sometimes solving problems you couldn't solve using earlier ML methods

Breakthroughs with neural networks

www.technewsworld.com/story/84013.html

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Microsoft AI Beats Humans at Speech Recognition

By Richard Adhikari
Oct 20, 2016 11:40 AM PT

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Image: Adobe Stock

Microsoft's Artificial Intelligence and Research Unit earlier this week reported that its speech recognition technology had surpassed the performance of human transcriptionists.

How do you feel about Black Friday and Cyber Monday?

- ☐ They're great -- I get a lot of bargains!
- ☐ The deals are too spread out -- I'd prefer just one day.
- ☐ They're a fun way to kick off the holiday season.
- ☐ I don't like the commercialization of Thanksgiving Day.
- ☐ They're crucial for the retail industry and the economy.
- ☐ The deals typically aren't that good.

Vote to See Results

E-Commerce Times

Black Friday Shoppers Hungry for New Experiences, New Tech

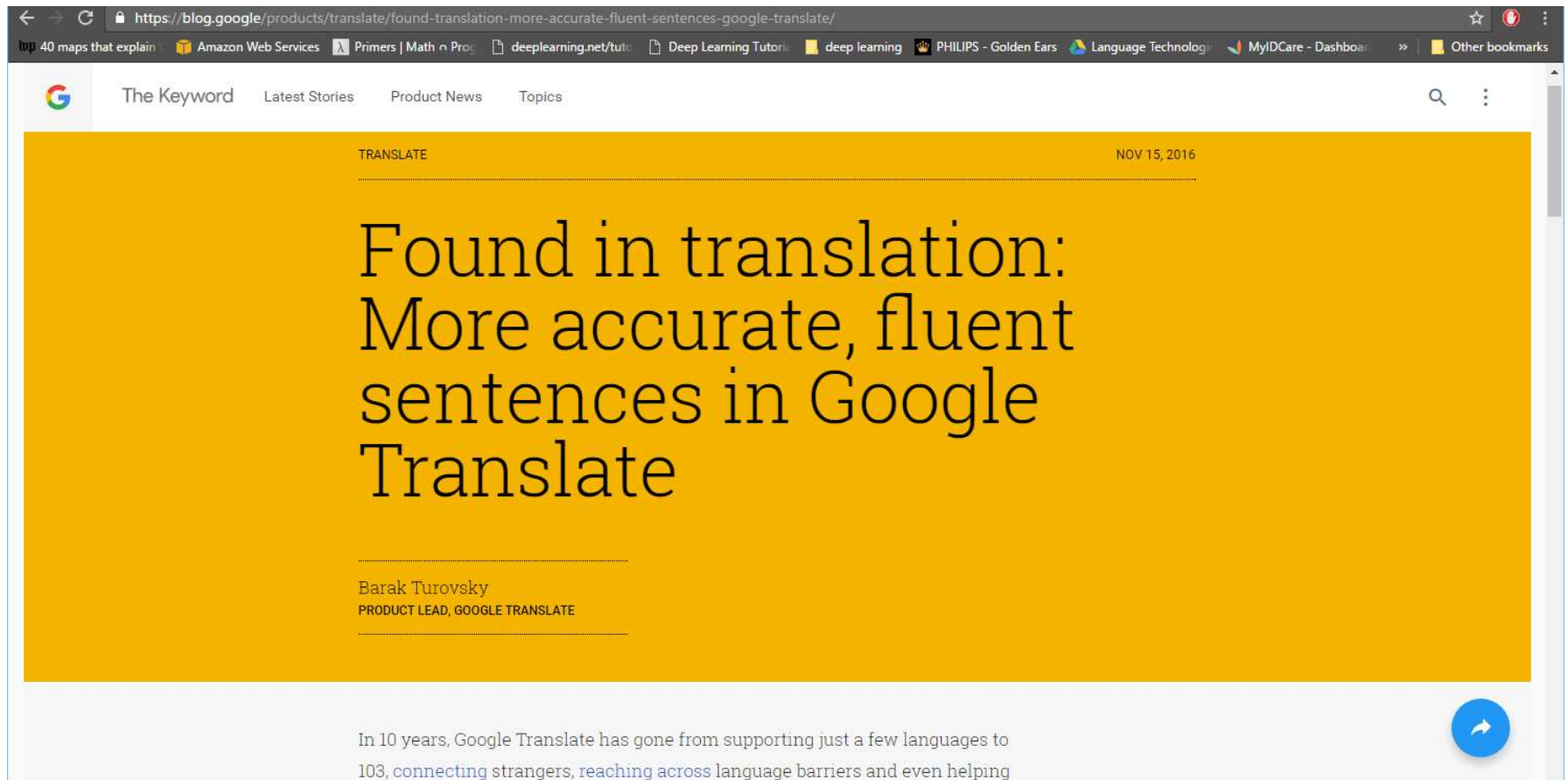
Pay TV's Newest Innovation: Giving Users Control

Apple Celebrates Itself in \$300 Coffee Table Tome

AWS Enjoys Top Perch in IaaS, PaaS Markets

US Comptroller Gears Up for Blockchain and

Breakthrough with neural networks



Breakthroughs with neural networks

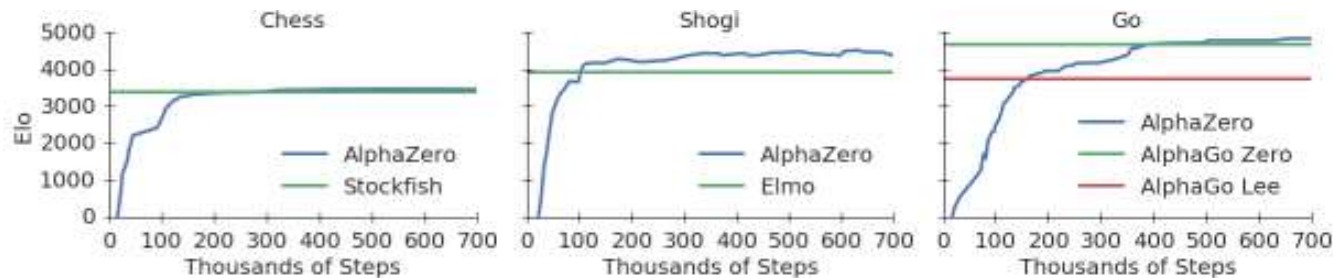
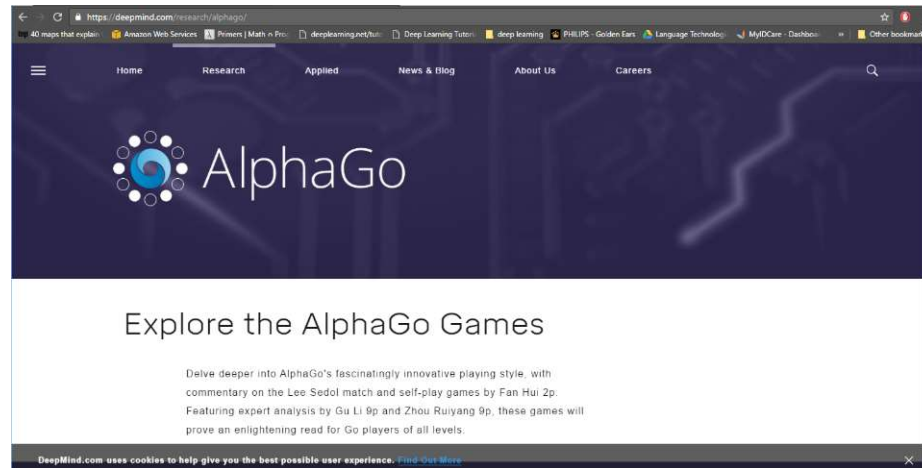
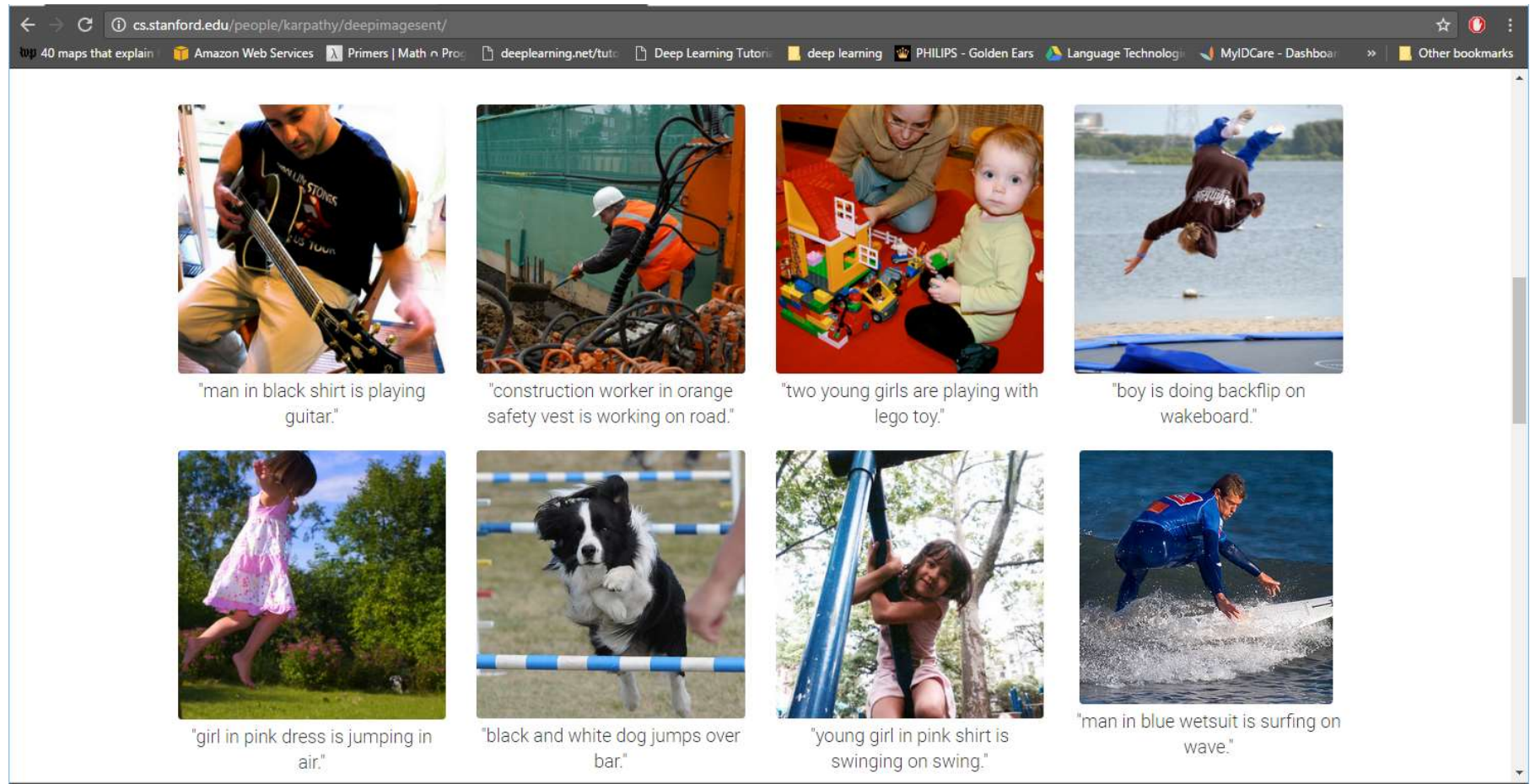


Figure 1: Training *AlphaZero* for 700,000 steps. Elo ratings were computed from evaluation games between different players when given one second per move. **a** Performance of *AlphaZero* in chess, compared to 2016 TCEC world-champion program *Stockfish*. **b** Performance of *AlphaZero* in shogi, compared to 2017 CSA world-champion program *Elmo*. **c** Performance of *AlphaZero* in Go, compared to *AlphaGo Lee* and *AlphaGo Zero* (20 block / 3 day) (29).

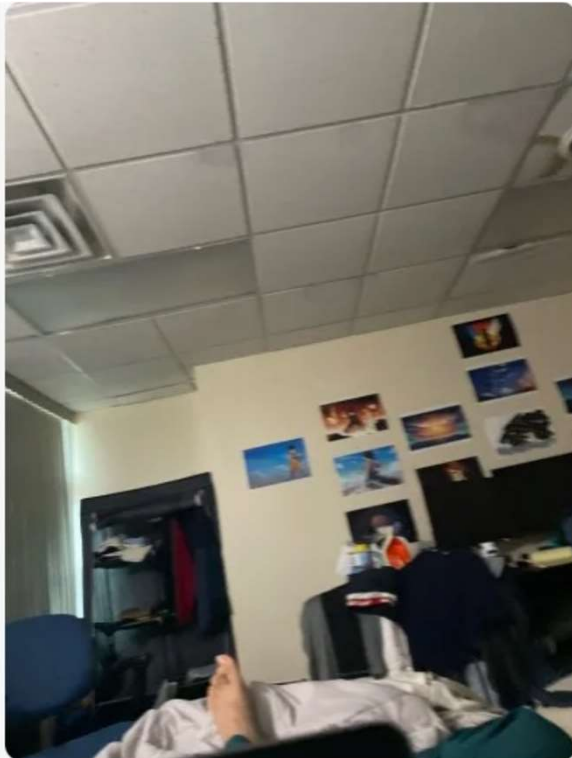
Success with neural networks



- Captions generated entirely by a neural network

And now...

ChatGPT, tell me about this picture



7:12 PM The ceiling in the photo appears to have several noticeable water stains, which could indicate potential water damage. The stains seem to be spread across multiple tiles, suggesting there might be a leak from the roof or pipes above. It would be advisable to address this issue promptly to prevent any further damage or potential mold growth. Notifying building maintenance or the landlord to investigate and fix the source of the leak would be the best course of action.

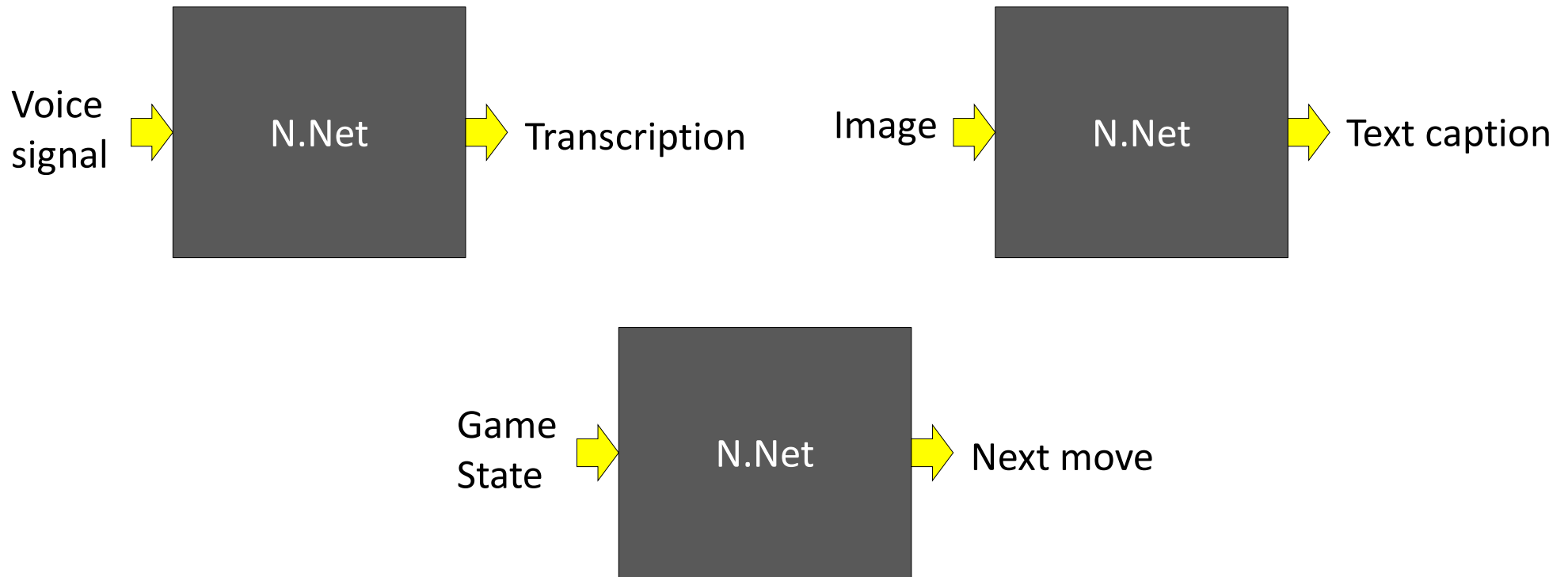
Besides the water stains on the ceiling, the photo shows a few other details:

1. **Posters on the Wall:** There are several posters or prints on the wall featuring various images, likely of scenes from movies, video games, or other forms of media.
2. **Furniture and Clutter:** The room contains a few pieces of furniture, including a blue chair, a black wardrobe or storage unit, and a desk with two monitors. There are clothes and other items visible on the wardrobe and desk, indicating a typical lived-in space.
3. **Lighting and Ventilation:** The ceiling has a light fixture and an air vent, suggesting this is a standard room with typical building amenities.
4. **Bed and Bedding:** The lower part of the photo shows a bed with a person lying on it, with their foot visible in the foreground. The bed has a comforter and some clothing or linens on it.
5. **Overall Condition:** The room appears somewhat cluttered, with personal belongings spread out. The presence of water stains on the ceiling is the most concerning issue visible.

Successes with neural networks

- And a variety of other problems:
 - From art to astronomy to healthcare..
 - and even predicting stock markets!

So, what are neural networks??



- What's in these boxes?

So, what are neural networks??



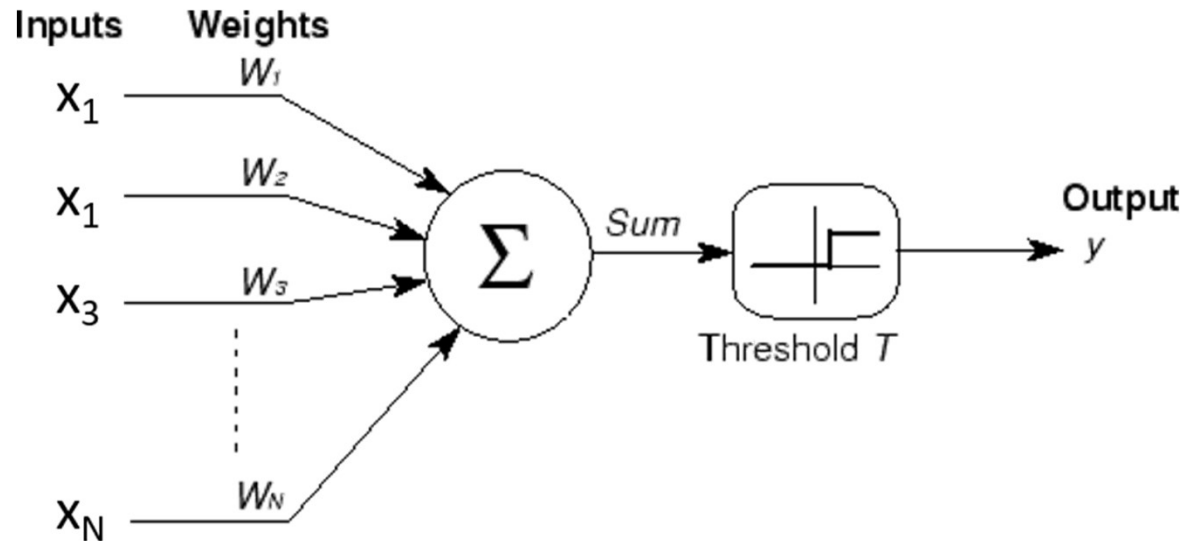
- It begins with this..

Brain: Interconnected Neurons



- Many neurons connect *in* to each neuron
- Each neuron connects *out* to many neurons
- The brain is a *network* of neurons

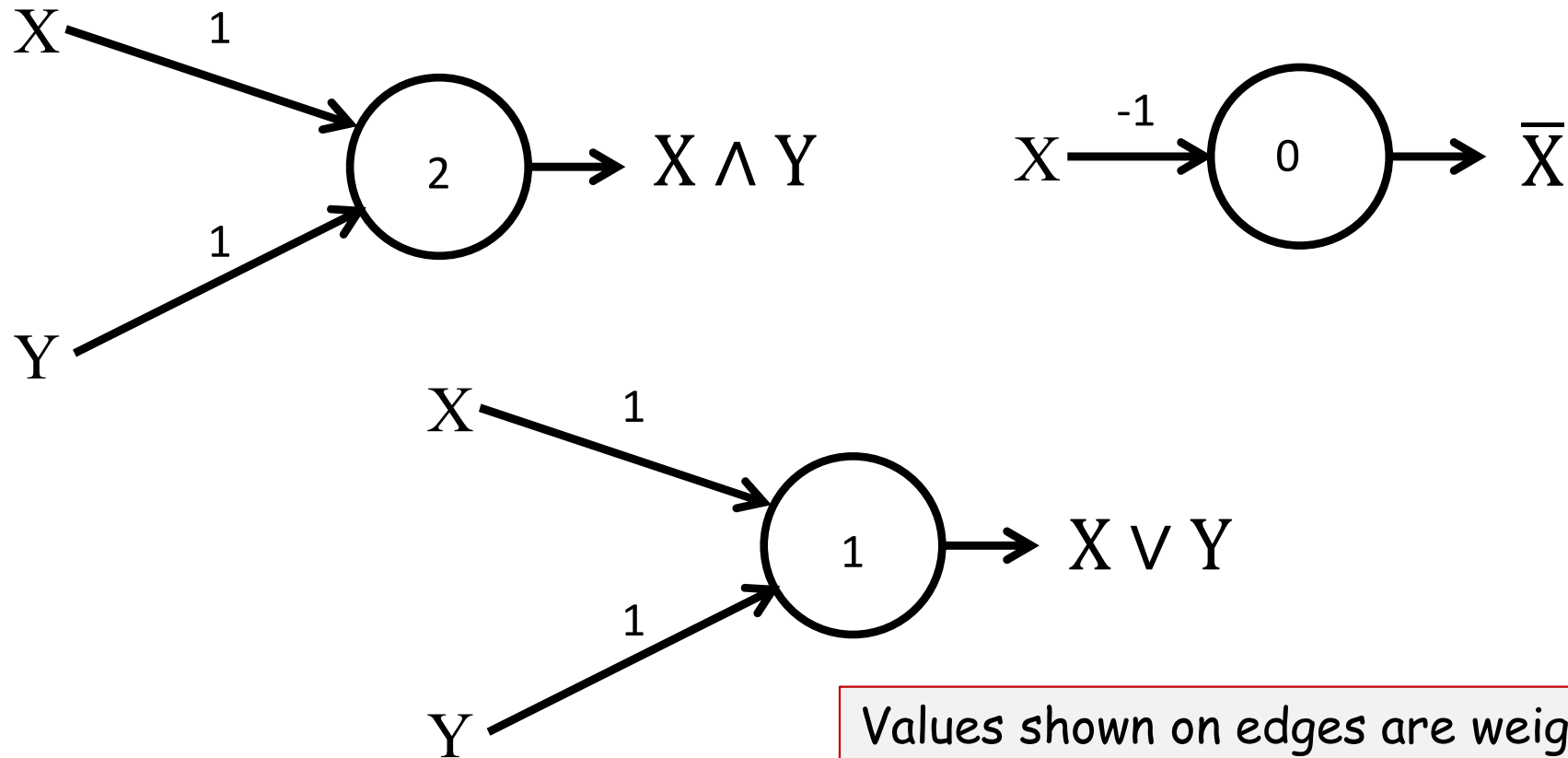
Perceptron: Simplified model



- Number of inputs combine linearly
 - Threshold logic: Fire if combined input exceeds threshold

$$Y = \begin{cases} 1 & \text{if } \sum_i w_i x_i - T \geq 0 \\ 0 & \text{else} \end{cases}$$

Perceptron

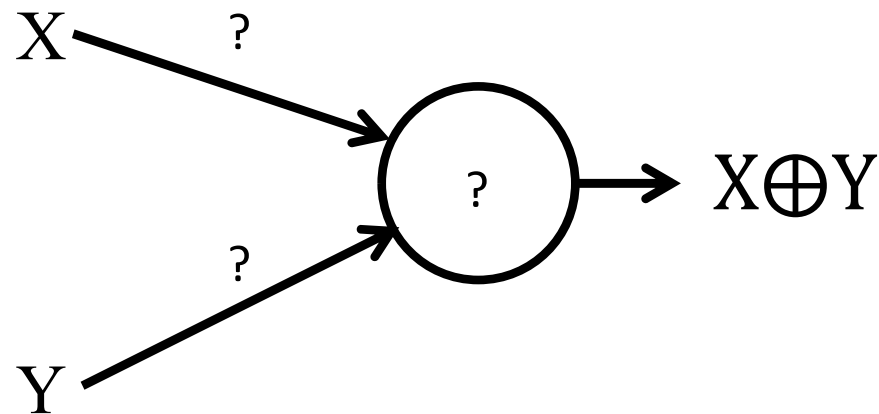


Values shown on edges are weights, numbers in the circles are thresholds

- Easily shown to mimic any Boolean gate
- But...

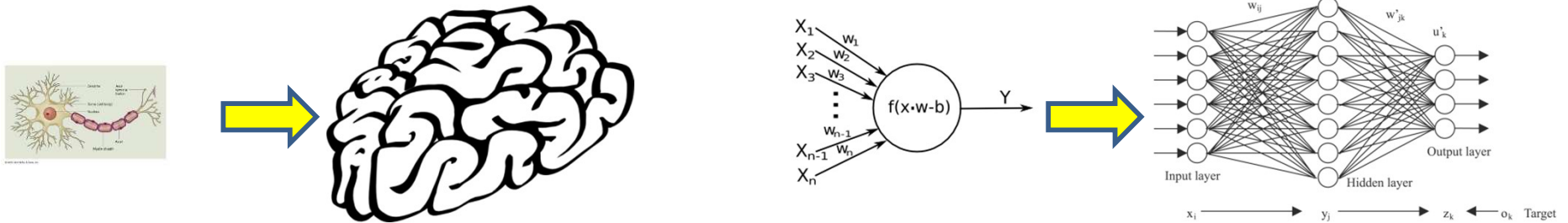
Individual units

No solution for XOR!



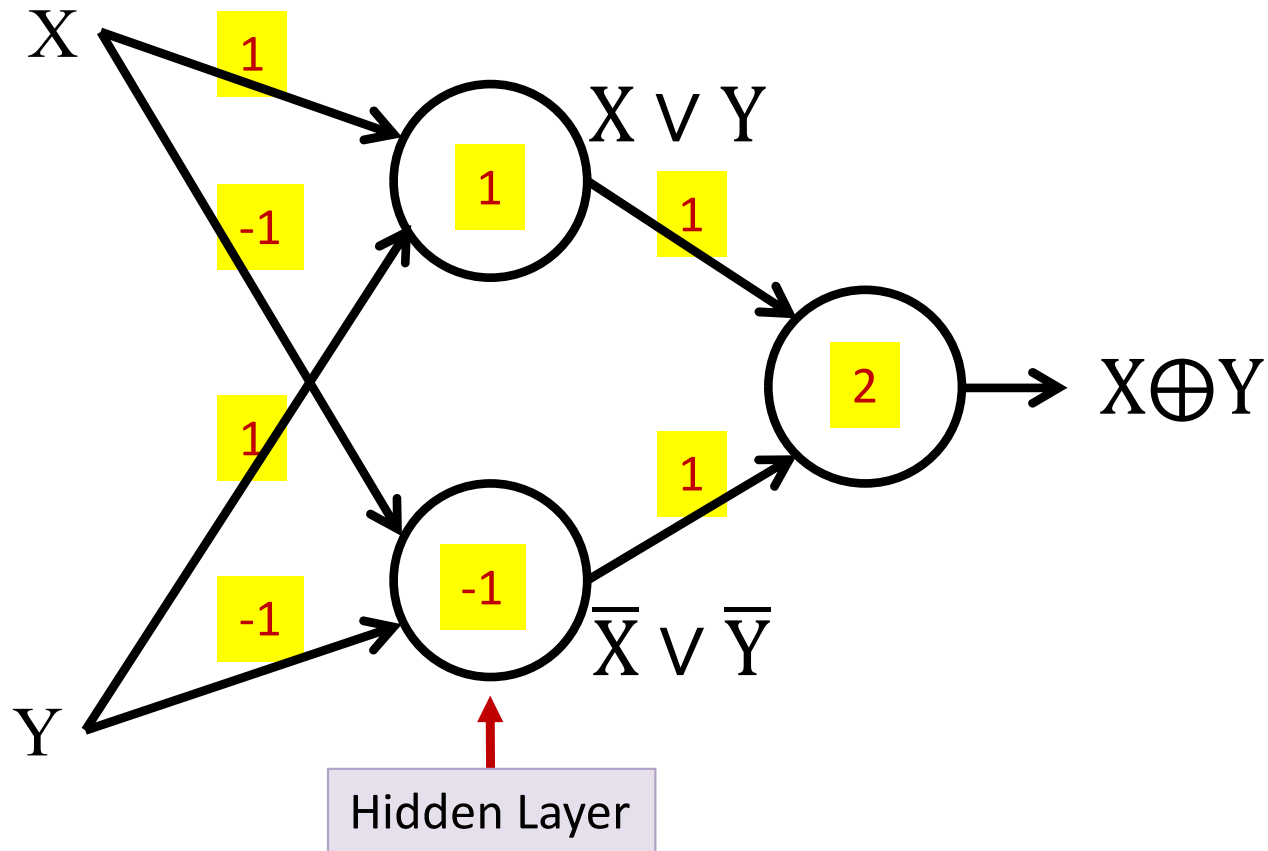
A perceptron is the simplest type of artificial neural network and is used for binary classification problems. It works like a decision-making unit that takes inputs, applies weights, and produces an output (0 or 1).

A single neuron is not enough



- Individual elements are weak computational elements
 - Marvin Minsky and Seymour Papert, 1969, *Perceptrons: An Introduction to Computational Geometry*
- *Networked* elements are required

Multi-layer Perceptron!



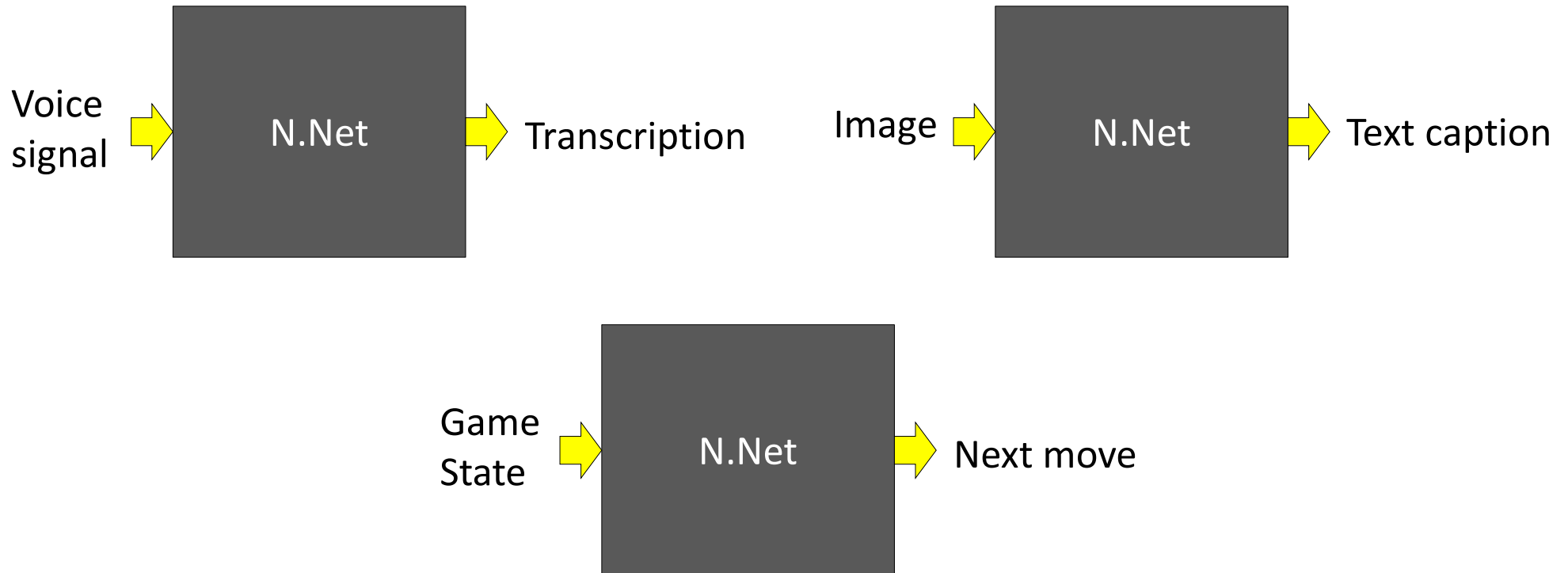
- **XOR**

- The first layer is a “hidden” layer

NNets in AI

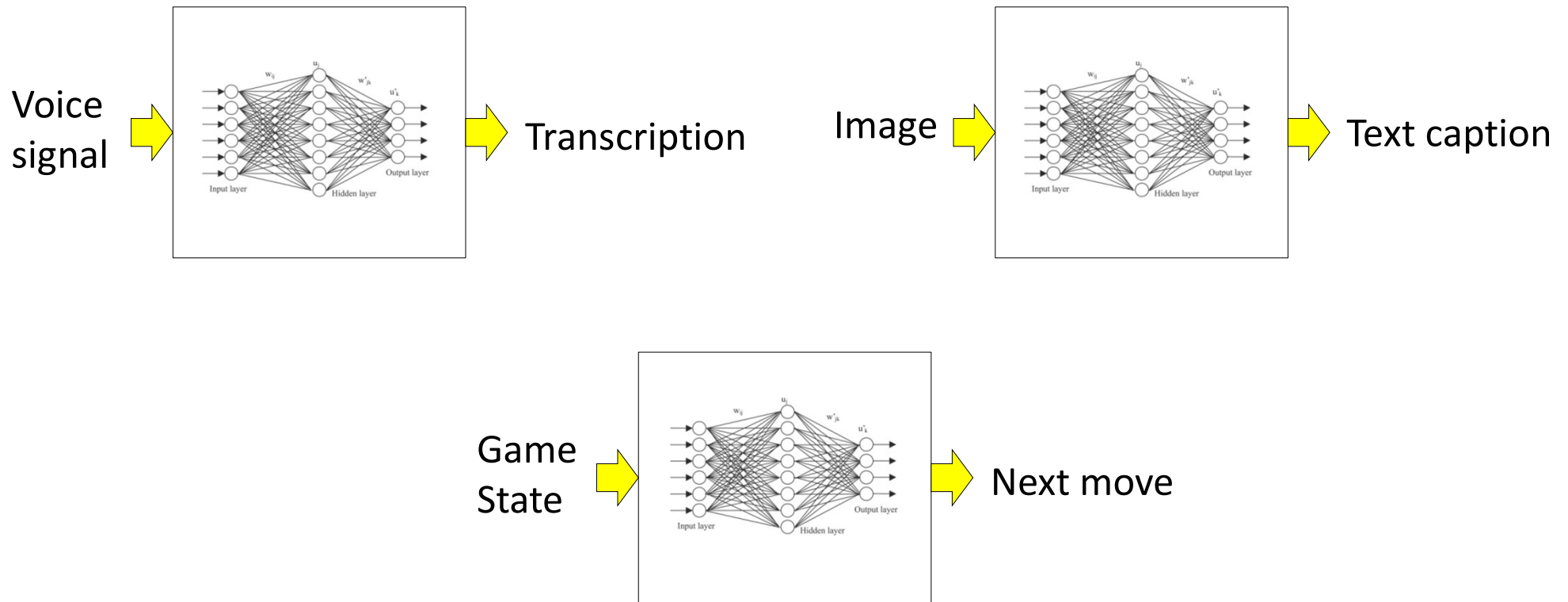
- The network is a function
 - Given an input, it computes the function layer wise to predict an output
 - More generally, given one or more inputs, predicts one or more outputs

These tasks are *functions*



- Each of these boxes is actually a function
 - E.g $f: \text{Image} \rightarrow \text{Caption}$

These tasks are *functions*



- Each box is actually a function
 - E.g $f: \text{Image} \rightarrow \text{Caption}$
 - It can be approximated by a neural network